

# Krishnasai Bairi

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## Professional Summary

Software Developer with 4 years of experience building autonomous AI agents, enterprise automation platforms, and scalable Python backend services. Proven track record designing LLM-driven orchestration agents that cut exception resolution time by over 95% and eliminate 80%+ of manual operational work. Skilled in Python, REST APIs, distributed databases, agentic workflows, and workflow automation. Currently building production-ready Generative AI applications using FastAPI, Retrieval-Augmented Generation (RAG), semantic search, embeddings, vector databases, and modern backend architecture.

## Technical Skills

**Languages:** Python, JavaScript, SQL

**Backend:** FastAPI, REST APIs, Pydantic, Async Python, gRPC, Protocol Buffers

**Generative AI & Agents:** LLMs, AI Agents, Agentic Workflows, Retrieval-Augmented Generation (RAG), Semantic Search, Embeddings, Prompt Engineering, LangChain, LangGraph

**Databases:** Google Cloud Spanner, SQL, Vector Databases

**Tools & Practices:** Git, Docker, Blaze, Google Apps Script, Agile, CI/CD

## Professional Experience

### Software Developer

Jul 2022 – Present

- Architected and deployed an autonomous, LLM-driven orchestration agent to triage and resolve enterprise transaction validation failures across 5+ error classes, reducing Mean Time to Resolution (MTTR) by over **95%** – from hours of manual work to under 5 minutes per transaction.
- Automated more than **80%** of recurring operational exceptions in high-volume enterprise workflows using AI-driven orchestration and intelligent business rule engines, eliminating manual engineer intervention.
- Built Python services querying distributed relational databases (Google Cloud Spanner) and Protocol Buffers/gRPC interfaces to run automated integrity checks across transaction logs, resource mappings, and catalog data.
- Programmed dynamic escalation workflows that detect partner and vendor capabilities and automatically route updates through integration APIs and automated communications, streamlining cross-company operations.
- Designed a high-coverage automated test suite with mocked database fixtures under a modern build system, ensuring zero-regression production deployments.
- Developed Python data pipelines streaming real-time metrics from internal APIs into reporting dashboards, cutting manual tracking effort by **5–8 hours per week** and improving SLA visibility.
- Designed SLA and ticket-aging algorithms with priority-based and working-day calculations that auto-exclude week-ends/holidays, achieving **100% calculation accuracy** and cutting report verification time by over **50%**.

*Key Initiative – Business Process Automation & Monitoring Suite (Python, Google Apps Script, JavaScript):*

- Engineered an automated alert-management system that monitors critical system emails, parses key data using regular expressions, and auto-generates/updates issue-tracker tickets with dynamically generated summary tables, saving **5–10 hours per week** of manual monitoring and triage.
- Built a data-driven error-reporting pipeline that queries enterprise data sources, clusters and categorizes recurring errors, and auto-generates linked tracking spreadsheets with detailed transaction data, saving **8–12 hours per week** and accelerating root-cause analysis.
- Maintained a central tracking system integrated with Google Sheets, eliminating redundant issue creation and manual spreadsheet updates.

## Projects

### Enterprise AI Knowledge Assistant

Python | FastAPI | LangChain | LangGraph

- Built an enterprise AI assistant capable of answering organization-specific queries using Retrieval-Augmented Generation (RAG).
- Implemented semantic search using embeddings and vector databases to improve knowledge retrieval accuracy.
- Developed REST APIs using FastAPI with Pydantic models for request validation and structured response handling.
- Designed modular AI workflows using LangGraph and integrated Large Language Models for contextual conversations.
- Implemented prompt engineering techniques that improved response quality while reducing hallucinations.
- Built a scalable backend architecture following dependency injection and clean architecture principles.

## Education

**Bachelor of Technology, Computer Science and Engineering**

SR Engineering College, Warangal

**2019 – 2022**

CGPA: **8.8/10**

## Certifications

Associate Data Practitioner – Google